

Retail Markdown Pricing and Margin Impact Analysis: Measuring Incremental Lift, Price Elasticity and the Profitability of Discounts

Your Name Project Link

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Abstract

Retailers mark products down to move stock and lift sales, and a markdown almost always sells more units. Whether those extra units pay for the price given up is a different question. This project studies 43,400 personal-care and beauty products, each taken through four sequential rounds of markdowns, to measure how much a discount actually adds and what it does to revenue and contribution margin. Three results stand out. First, measuring lift against the original baseline badly misattributes the effect of later markdowns; lift has to be measured against the period immediately before each markdown. Second, demand is price-inelastic (median elasticity -0.65), and the response depends almost entirely on whether a category is in season: in-season products respond about 5.6 times more strongly to the same discount. Third, over the full markdown cycle revenue rose 10.7% relative to a no-markdown counterfactual, while contribution margin fell 22.9% under an assumed 40% cost of goods. A simulation of alternative policies suggests stopping off-season markdowns and capping in-season discounts at around 10%. The problem framing is adapted from Increff's markdown optimization case study; the data is synthetic.

Keywords: markdown optimization, price elasticity, incremental lift, ANOVA, Tukey HSD, counterfactual analysis, contribution margin

1 Introduction

Markdowns are one of the most frequently used and least carefully measured levers in retail. A discount that moves more units looks successful on a sales report, yet it may simply transfer margin to customers who would have bought anyway. The decision has three parts: *which* products to discount, *how deep* to go, and *when*. Getting it wrong in one direction leaves unsold stock; in the other, it gives away profit.

This project was inspired by Increff's case study on markdown optimization [1], which frames exactly this decision for a retailer. The aim was to practise the analysis that sits behind such a decision, end to end, on a public dataset. The work is organised around four questions:

- Q1.** How much additional volume does a markdown actually generate, measured correctly?
- Q2.** How sensitive is demand to price, and does that sensitivity vary by season, category or promotion channel?
- Q3.** Do deeper discount tiers perform significantly differently from shallower ones?
- Q4.** Taking cost into account, did the markdown programme create or destroy value, and what policy would have done better?

The same logic applies well beyond apparel and beauty retail. Any business that runs coupons or promotions, including food delivery and quick commerce, faces the question of whether an incentive generates *incremental* demand or subsidises demand that already existed.

2 Data and Preprocessing

2.1 Dataset

The analysis uses the *Retail Markdown Optimization: Discounts & Sales* dataset [2], a synthetic dataset modelled on a French personal-care and beauty retailer. Each row is a product, observed at a pre-markdown baseline and after each of four sequential markdowns. Table 1 summarises the fields used.

Table 1: Dataset overview (after cleaning)

Field	Description
Products	43,400 unique products (43,750 raw rows)
Category	Skincare (19,530), Bodycare (13,020), Makeup (10,850)
Season, Brand, Channel	4 seasons; 6 brands; promotion channel: in-store, online, social media
Prices	Original price 5–100 (mean 52.7); competitor price (mean 53.7)
Seasonality factor	Score from 0.5 to 2.0 (mean 1.33)
Baseline sales	Units sold before any markdown (mean 171.9, range 26–416)
Markdowns	Four rounds per product: 10–40%, 20–50%, 10–30%, 30–50%
Sales after markdown	Units sold after each of the four rounds
Not available	Product cost, calendar dates, stock figures consistent with sales

2.2 Data quality

Three issues were found and handled before any modelling:

- **Duplicate records.** 350 rows were exact duplicates of other rows (same product ID and identical values in every column). They were removed, leaving 43,400 products.
- **A leaking column.** The dataset provides an “Optimal Discount” field. Its correlation with the mean of each product’s four markdowns is 0.997, so it is derived from the inputs rather than being an independent benchmark. It was excluded; treating it as a target would amount to grading the markdowns against themselves.
- **No cost data.** Contribution margin requires cost of goods sold (COGS). COGS was therefore set as an assumption (40% of original price) and tested at 30% and 50%.

2.3 Reshaping to event level

The four markdown and sales columns were unpivoted in SQL (DuckDB [6]) into one row per product and markdown round, giving 173,600 markdown events. A `LAG` window function, partitioned by product and ordered by round, attached the previous period’s sales to each event (Figure 1). The full SQL is linked under Code Availability.

Figure 2 shows the design of the four rounds and the distribution of lift. Rounds 1 and 3 are shallow, rounds 2 and 4 deep. Incremental lift has a mean of 17.9% and a median of 16.7%, with mild right skew (skewness 0.40).

3 Methodology

Figure 3 summarises the analysis. The key definitions are set out below.

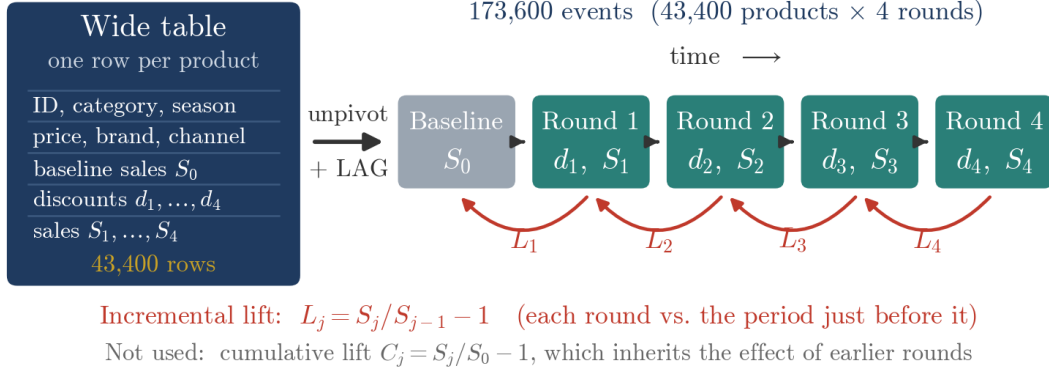


Figure 1: Reshaping the wide product table into markdown events and defining incremental lift.

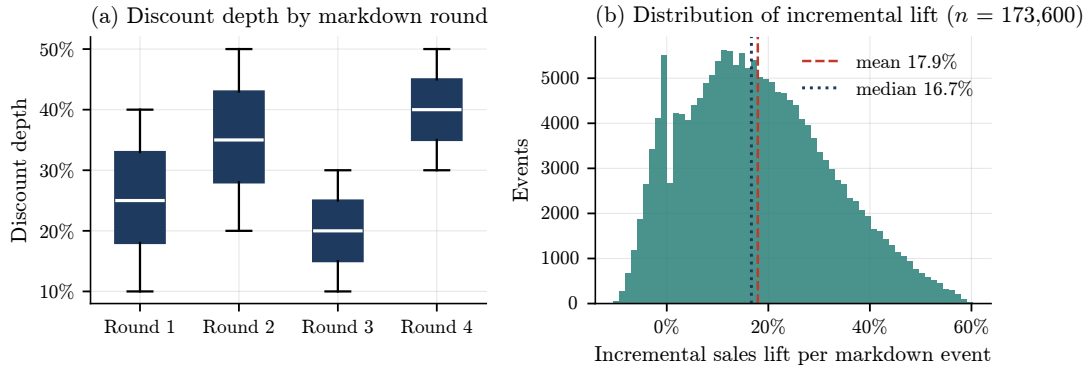


Figure 2: (a) Discount depth by round. (b) Distribution of incremental lift across all 173,600 markdown events.

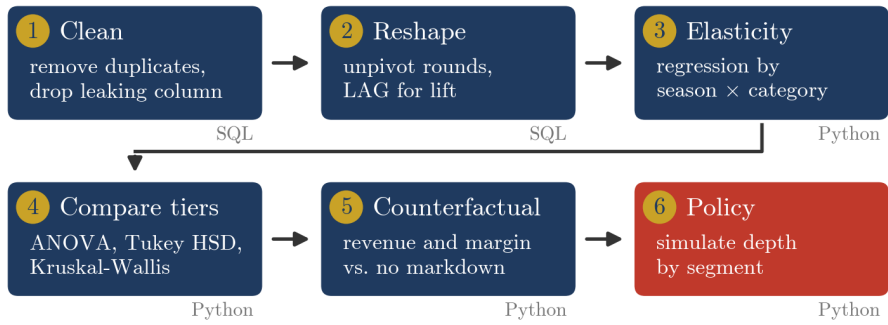


Figure 3: Analysis pipeline. Steps 1–2 run in SQL (DuckDB); steps 3–6 in Python (pandas, statsmodels, SciPy).

3.1 Incremental lift and break-even

For product i in round $j \in \{1, \dots, 4\}$ with discount d_{ij} and unit sales S_{ij} (S_{i0} being the baseline), incremental and cumulative lift are

$$L_{ij} = \frac{S_{ij}}{S_{i,j-1}} - 1, \quad C_{ij} = \frac{S_{ij}}{S_{i0}} - 1. \quad (1)$$

Because sales in this data carry forward from one round to the next, C_{ij} mixes the effect of round j with every earlier round. All analysis therefore uses L_{ij} . A markdown holds period revenue constant only if $(1 - d)(1 + L) \geq 1$, i.e. if lift reaches the break-even level

$$L^*(d) = \frac{d}{1 - d}, \quad (2)$$

so a 30% discount needs 43% more units just to stand still on revenue.

3.2 Price elasticity and the response model

Event-level elasticity is $\varepsilon_{ij} = -L_{ij}/d_{ij}$, and medians are reported. Because lift turned out to be close to proportional to depth, the response is modelled without an intercept:

$$L_{ij} = d_{ij} (\beta_{s(i)} + \gamma^\top \mathbf{x}_i) + \delta_j + u_{ij}, \quad (3)$$

where $\beta_{s(i)}$ is the response slope for product i 's season-category segment, $\mathbf{x}_i$ contains promotion channel, brand and the price gap to competitors, and δ_j are round effects. Four specifications were compared (M1: depth only; M2: depth $\times$ seasonality factor; M3: depth $\times$ season $\times$ category; M4: M3 plus the seasonality factor). Each product contributes four correlated events, so standard errors are clustered by product [5].

3.3 Comparing discount tiers

Events were grouped into four depth tiers (10–20%, 20–30%, 30–40%, 40–50%). A one-way ANOVA assumes independent observations, which the four rounds of the same product are not. To respect this, **one round was drawn at random from each product**, giving 43,400 independent observations. Differences were tested with a one-way ANOVA (effect size η^2), checked with the non-parametric Kruskal–Wallis test [4], and followed by Tukey's HSD [3] for all six pairwise comparisons at a family-wise error rate of 5%.

3.4 Counterfactual revenue and margin

Actual outcomes over the four rounds were compared with a counterfactual in which every product stays at full price and baseline volume. With original price P_i and COGS fraction c :

$$R_i^{\text{act}} = \sum_{j=1}^4 P_i(1 - d_{ij})S_{ij}, \quad R_i^{\text{cf}} = 4P_iS_{i0}, \quad (4)$$

$$M_i^{\text{act}} = \sum_{j=1}^4 P_i(1 - d_{ij} - c)S_{ij}, \quad M_i^{\text{cf}} = 4P_i(1 - c)S_{i0}. \quad (5)$$

3.5 Policy simulation

For each segment s , a uniform depth $d \in \{10\%, 15\%, \dots, 50\%\}$ was held for four rounds, with sales compounding as $S_j = S_0(1 + \hat{\beta}_s d)^j$. Revenue and margin were compared with the counterfactual, and the revenue-maximising and margin-maximising depths recorded. Depths below 10% were not simulated because the data contains none.

4 Results

4.1 Cumulative measurement misattributes lift

Table 2 and Figure 4 contrast the two ways of measuring lift. Under the cumulative view, round 3 (mean depth 20%) appears to outperform round 2 (35%), 58.3% against 40.0%. That result is an artefact: round 3 inherits the growth of the two rounds before it. Measured incrementally, round 3 added 11.9% and round 2 added 20.9%, the ordering one would expect from their depths. A team working from the cumulative numbers would favour exactly the wrong markdowns.

Table 2: Markdown rounds: depth, incremental and cumulative lift, and break-even lift (computed in SQL)

Round	Depth range	Mean depth	Incremental lift		Cumulative lift	Break-even lift
			mean	median		
1	10–40%	25.1%	15.0%	14.0%	15.0%	35.3%
2	20–50%	35.0%	20.9%	21.2%	40.0%	56.7%
3	10–30%	20.0%	11.9%	11.7%	58.3%	25.6%
4	30–50%	40.0%	23.9%	25.5%	100.8%	68.3%

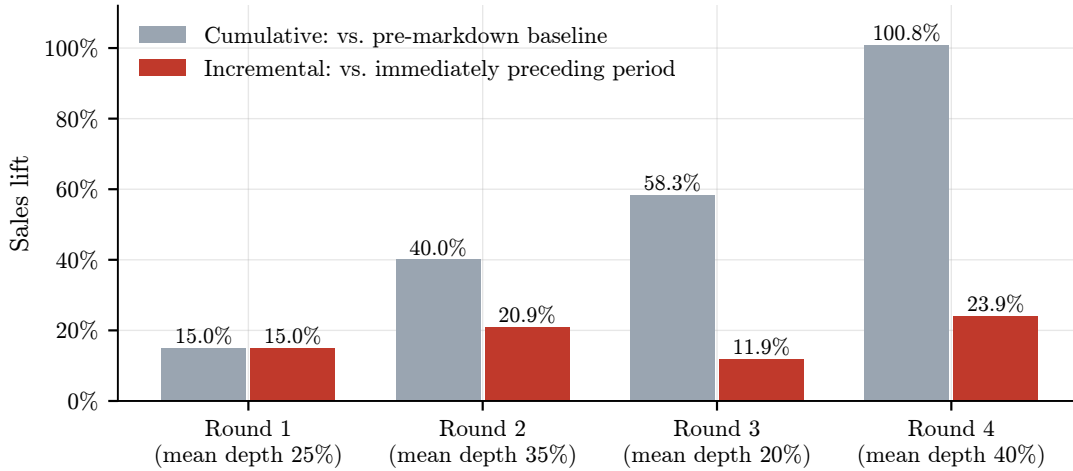


Figure 4: Cumulative versus incremental lift by round.

4.2 Demand response depends on season fit, not on the seasonality score

Lift rises almost linearly with depth, so the useful quantity is the slope: how many points of lift each point of discount buys. Depth alone explains 21% of the variation in lift (Table 3). Letting the slope depend on the dataset’s seasonality factor raises this to 49%, and the factor’s coefficient is large and highly significant (0.497). Letting the slope depend on season and category instead raises it to 79%.

Table 3: Response models for incremental lift (no intercept; standard errors clustered by product)

Model	Depth interacted with	R^2	Seas. factor coef.
M1	nothing	0.212	–
M2	seasonality factor + controls	0.486	0.497 ($p < 0.001$)
M3	season $\times$ category + controls	0.791	–
M4	M3 + seasonality factor	0.791	0.0002 ($p = 0.95$)

M4 settles what drives the response. Once season and category are in the model, the seasonality factor has no effect at all ($p = 0.95$). It looked important in M2 only because it is high for most in-season combinations, so it was acting as a proxy for the real driver. The estimated slopes fall into two clear groups (Figure 5): eight in-season segments with a slope of about 0.85, and four off-season segments (Makeup and Skincare in the rainy season; Bodycare and Skincare in Spring) with a slope of about 0.15. The same discount therefore buys roughly 5.6 times more lift when a category is in season. None of the control variables matter: promotion channel ($p = 0.74$ and 0.46), the price gap to competitors ($p = 0.78$) and round ($p > 0.77$) are all insignificant.

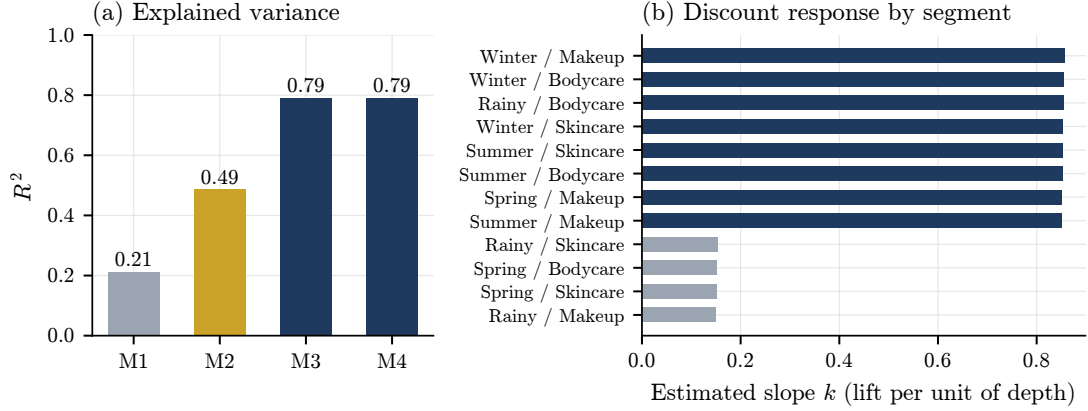


Figure 5: (a) Explained variance of models M1–M4. (b) Estimated response slope $\hat{\beta}_s$ for each season–category segment.

In elasticity terms, demand is inelastic everywhere: the median across all events is -0.65 , in-season segments sit at -0.85 and off-season segments at -0.15 (Figure 6). The marginal medians in panel (b) are averages over this pattern. Skincare’s lower figure (-0.50), for instance, only reflects that it is off season in two of the four seasons; in season it responds like any other category.

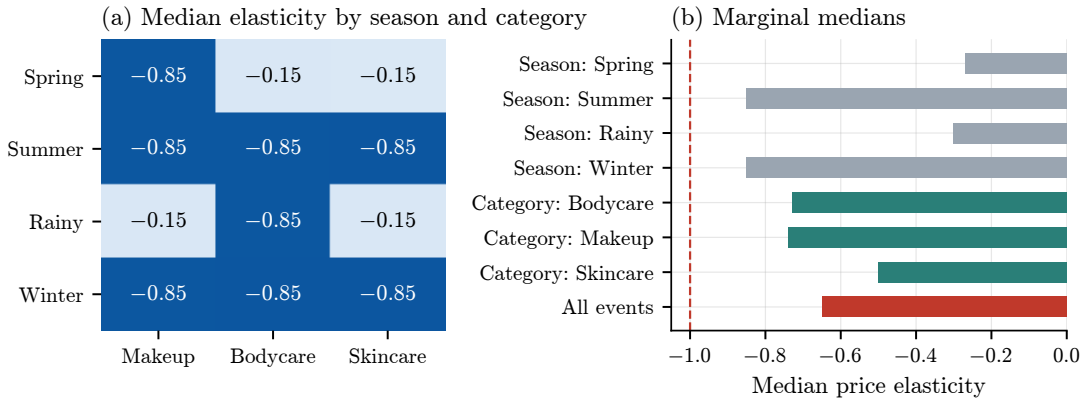


Figure 6: (a) Median price elasticity by season and category. (b) Marginal medians by season, by category and overall.

Figure 7 places both groups against the break-even curve. Neither comes close at any depth, which is why only 0.6% of all markdown events (0.97% in season, none off season) recovered their own discount within the period.

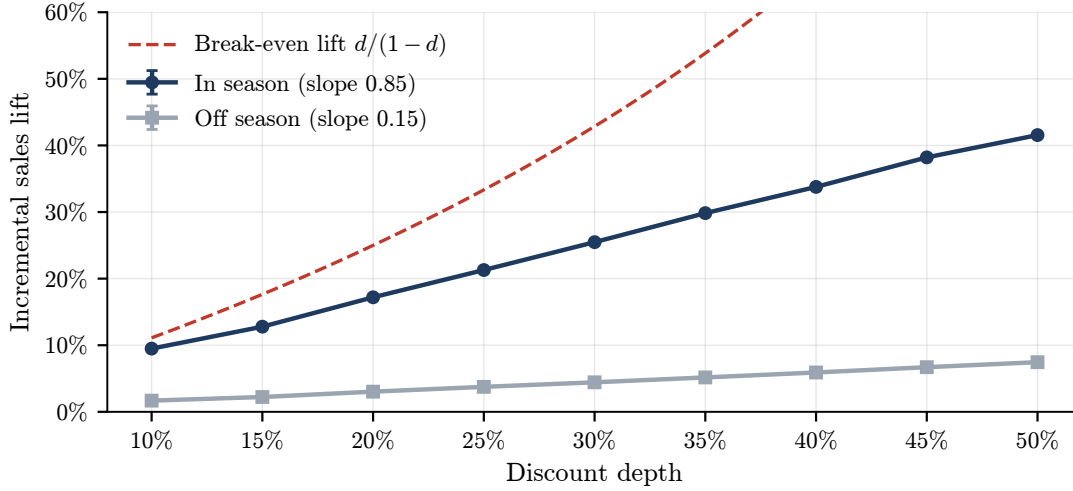


Figure 7: Incremental lift by discount depth for in-season and off-season products (95% CIs), against the break-even lift $d/(1-d)$.

4.3 Discount tiers differ significantly, and deeper tiers lose more revenue

The four tiers differ significantly in lift: $F = 3,434$, $p < 0.001$, $\eta^2 = 0.19$. The Kruskal–Wallis test agrees ($H = 7,678$, $p < 0.001$). Tukey’s HSD finds every one of the six pairs significantly different, both for unit lift and for period revenue (Table 5, Figure 8). Each 10-point step in depth adds about 6 points of lift, and each step also loses revenue for the period, from -7.2% in the shallowest tier to -30.0% in the deepest (Table 4). There is no sweet spot where a deeper cut starts to pay.

Table 4: Outcomes by discount tier (one randomly selected round per product)

Tier	n	Unit lift	95% CI	Period revenue change
10–20%	8,653	8.9%	± 0.1 pp	-7.2%
20–30%	12,402	14.7%	± 0.2 pp	-13.5%
30–40%	12,749	20.6%	± 0.2 pp	-21.1%
40–50%	9,596	26.6%	± 0.4 pp	-30.0%

Table 5: Tukey HSD pairwise comparisons of unit lift (percentage points)

Comparison	Mean difference	95% family-wise CI	Adjusted p
20–30% vs 10–20%	5.85	[5.40, 6.31]	< 0.001
30–40% vs 10–20%	11.71	[11.26, 12.16]	< 0.001
40–50% vs 10–20%	17.71	[17.22, 18.19]	< 0.001
30–40% vs 20–30%	5.86	[5.45, 6.27]	< 0.001
40–50% vs 20–30%	11.85	[11.41, 12.29]	< 0.001
40–50% vs 30–40%	6.00	[5.56, 6.43]	< 0.001

4.4 Revenue rose, contribution margin fell

Against the no-markdown counterfactual, the extra volume was worth 61.1 index points of revenue at full price and the discounts gave back 50.4, leaving revenue 10.7% higher (Figure 9). Margin behaves differently: each extra unit carries its own cost, while the discount comes directly out of profit. Contribution margin ended 22.9% below the counterfactual, and 97.9% of products were worse off.

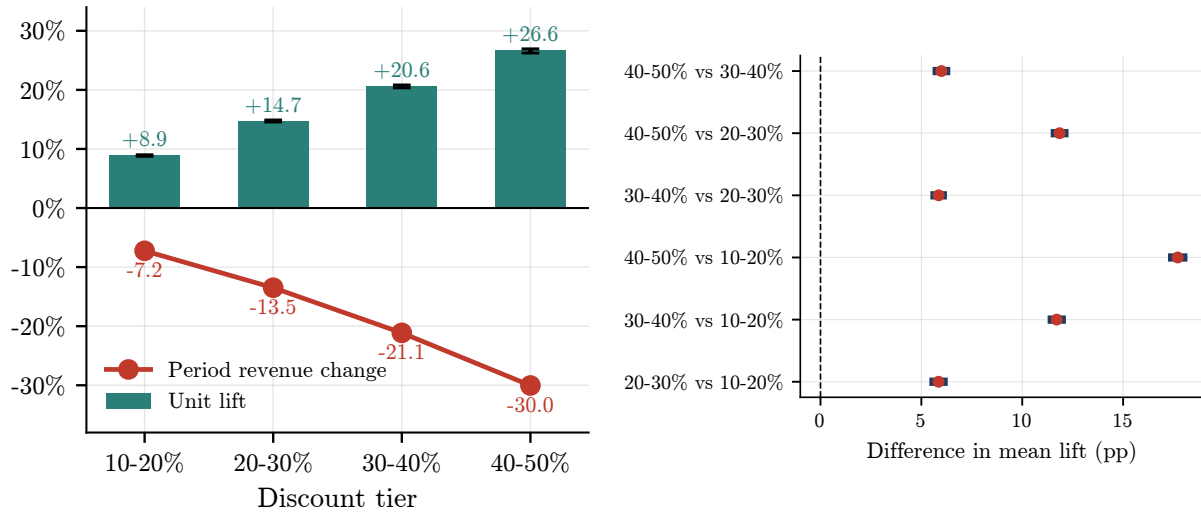


Figure 8: Left: unit lift (bars, 95% CI) and period revenue change (line) by tier. Right: Tukey HSD intervals; none crosses zero.

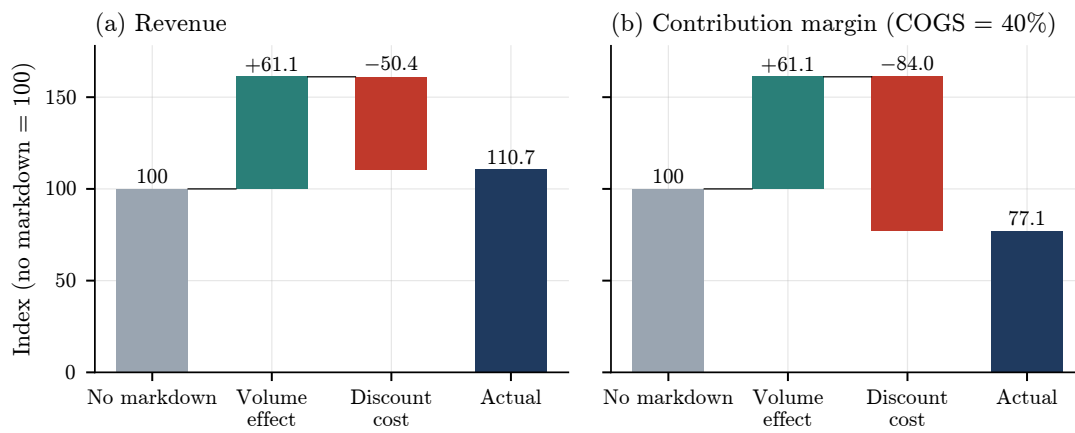


Figure 9: Decomposition of revenue and contribution margin over the four rounds (no markdown = 100).

The conclusion does not depend on the cost assumption (Table 6): margin falls at every COGS level tested. Splitting by season fit sharpens it further. In-season products gained revenue but still lost margin; off-season products lost on both.

Table 6: Change vs. the no-markdown counterfactual: sensitivity to COGS, and split by season fit (COGS = 40%; computed in SQL)

Scenario	Revenue	Contribution margin	Products losing margin
COGS 30%	+10.7%	−10.9%	76.6%
COGS 40%	+10.7%	−22.9%	97.9%
COGS 50%	+10.7%	−39.7%	100.0%
In-season products	+21.6%	−15.7%	—
Off-season products	−22.5%	−44.9%	—

4.5 Policy simulation: revenue and margin point in opposite directions

Figure 10 shows the simulated outcome of holding one depth for all four rounds. For in-season segments, revenue keeps rising up to 45% off, reaching about 32% above the counterfactual. A team that sets discounts to maximise revenue or GMV would choose that depth, and would lose about 40% of contribution margin in the process. Margin is highest at 10% (about +2.7%) and turns negative a little past 20%. For off-season segments, every depth loses both revenue and margin; the least damaging option, 10%, still costs 13.5% of margin.

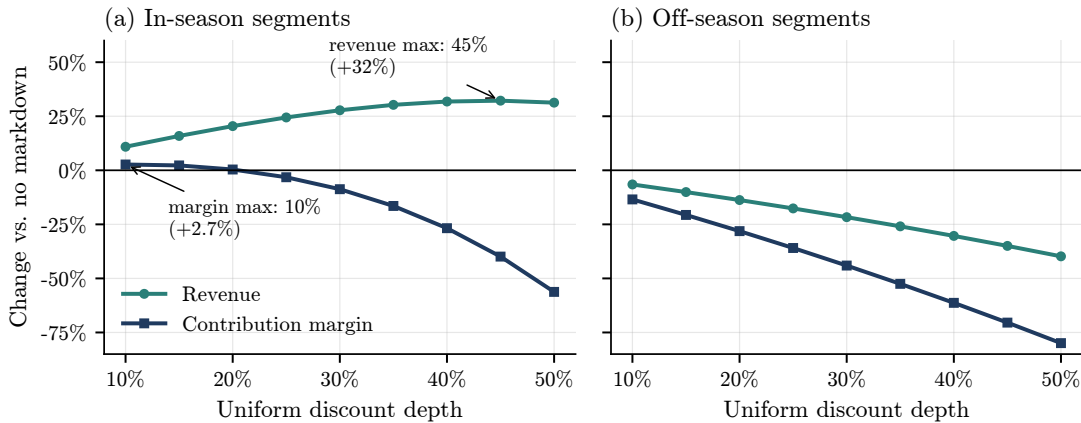


Figure 10: Simulated revenue and contribution margin against a uniform discount depth (COGS = 40%).

5 Discussion and Recommendations

The evidence points to a small number of concrete changes (Table 7). The broader lesson is about measurement: this markdown programme looked successful on units and revenue, and was loss-making on the metric that matters. Discount decisions should be judged on incremental units and contribution margin, with revenue as a supporting number.

One qualification matters. Markdowns also exist to clear stock that would otherwise be written off. This analysis gives unsold stock no cost, which favours shallow discounts. For aged or end-of-season inventory, a deeper markdown can still be the right decision, and judging that would require holding costs and salvage values.

Table 7: Recommendations

Recommendation	Evidence
Stop marking down off-season products	Revenue -22.5% and margin -44.9% vs. no markdown; no depth breaks even
Cap in-season markdowns near 10% (current average: 30%)	10% was the margin-maximising depth in all eight in-season segments
Do not set depth by revenue	Revenue peaks at 45% off, where margin is about 40% below the counterfactual
Stop paying a premium for any one promotion channel	No difference between in-store, online and social media
Report markdowns on incremental lift and margin	Cumulative lift mis-ranked the markdown rounds

6 Limitations and Future Work

- **Synthetic data.** The patterns reflect how the data was generated; the sharp two-group structure in Figure 5 is an example. The method, not the specific numbers, is what transfers to real data.
- **No undiscounted control.** Every product was discounted in every round, so a general trend cannot be fully separated from the discount effect. Results should be read as strong associations, not proven causal effects.
- **Assumed cost.** COGS was not observed, hence the sensitivity analysis.
- **Inventory not modelled.** The stock column did not reconcile with sales and was not used.
- **Limited depth range.** The data contains no discounts below 10%, so shallower policies cannot be evaluated.

With real transaction data, the natural next steps are a randomised holdout of products kept at full price to measure lift causally; adding cost, stock age and sell-through so that margin can be traded against the cost of unsold inventory; and estimating elasticity at the product level to set markdowns per SKU.

7 Conclusion

This project measured what a four-round markdown programme actually achieved for 43,400 products. Correctly measured, discounts added volume, but demand was inelastic (median elasticity -0.65) and the response depended mainly on whether a category was in season. All four discount tiers differed significantly, and each deeper tier lost more revenue than the last. Over the full cycle, revenue rose 10.7% while contribution margin fell 22.9%. The recommended policy is to stop off-season markdowns and to cap in-season discounts near 10%, and to judge every markdown on incremental units and margin rather than on sales or revenue.

Tools and Environment

- **SQL:** DuckDB, for auditing, de-duplication, unpivoting, CTEs, window functions and ROLLUP aggregation (script linked below).
- **Python:** pandas, NumPy, statsmodels [7] (OLS with cluster-robust errors, Tukey HSD), SciPy (ANOVA, Kruskal–Wallis), matplotlib.

- **Validation:** random one-event-per-product sampling for ANOVA, non-parametric cross-check, COGS sensitivity analysis.

Code Availability

The complete SQL pipeline (DuckDB) is available here: `markdown_pricing_analysis.sql`. It runs end to end from the raw CSV and covers the data audit and de-duplication, the unpivot of the four markdown rounds, incremental lift with `LAG`, the round-level comparison in Table 2, and the season-fit counterfactual in Table 6. The expected output is noted under each query.

References

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